Directional absorption asymmetry: the established two-port optimum and a sharp material-constrained thickness asymptotic

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Abstract

The requested fixed-transmission, fixed-mean-absorption optimization for a passive reciprocal two-port is established: it is a direct slice of the complete reflection-feasibility theorem of Guo and Fan (2022). We give its exact feasible region, optimum, and all equality cases. We then study a strictly narrower Maxwell problem: finite nonmagnetic dielectric multilayers at one frequency, with $1 \le \operatorname{Re}\varepsilon \le P$, $0 \le \operatorname{Im}\varepsilon \le Q$, fixed integrated loss M, and thickness at most L. In the strict active branch of the matrix optimum, we prove an exact inverse-square asymptotic for its unavoidable absorption-asymmetry deficit, including the optimal leading constant and exact fixed transmission and mean absorption. The lower bound is uniform over arbitrary finite layer counts; the upper bound is realized asymptotically by ordinary dielectric layers. Two independent mathematical agent reviews and independent transfer calculations found no obstruction. **Verdict: KNOWN** for the matrix problem and the two-port invariant; **CONDITIONAL** for the priority of the proved material-constrained asymptotic; **UNRESOLVED** for general three-dimensional OQ7 and finite-bandwidth extensions. No Lean kernel verification was obtained.

1 Novelty check before theorem development

The starting question is OQ7 in Section 4.1 of Miller and Monticone's tutorial [1]. A two-port specialization does not by itself solve the general optical cross-section question. Throughout this note, absorption and transmission are power fractions for a pair of power-normalized ports. For reciprocal two-port systems,

$$A_L - A_R = |r_R|^2 - |r_L|^2.$$

Thus the same difference measures the opposite imbalance in reflected power. For an extended free-space object, identifying all scattered power with one reflection channel would require additional assumptions.

1.1 Mandatory comparisons

Prior result	Statement and assumptions	Consequence for this investigation
Sounas–Alù (2014) [2]	Reciprocity and the optical theorem give $\sigma_{\text{ext}}(\mathbf{e},\mathbf{k}) = \sigma_{\text{ext}}(\mathbf{e}^*,-\mathbf{k})$. The polarization is conjugated when propagation reverses.	Extinction symmetry is established. It permits absorption and scattering to exchange their directional contributions; it does not specify a material and thickness budget.
Sounas–Alù (2017) [3]	For a lossless time-reversal-symmetric two-port, the pointwise ratio of oppositely excited internal intensities lies between $(1-\lvert r\rvert)/(1+\lvert r\rvert)$ and its reciprocal. A multiport radiation-efficiency inequality incorporates loss channels.	Internal-field asymmetry constraints are established. Applying the lossless two-port formula unchanged to an arbitrary lossy structure is invalid.
Guo–Fan (2022) [4]	Necessary and sufficient reflection feasibility at prescribed singular values, Eqs. (24)–(25). For ordered reflection magnitudes $p \geq q$ and singular values $s_1 \geq s_2$: $p-q \leq s_1-s_2 \leq p+q \leq s_1+s_2$.	Together with $s_1 \leq 1$ and the Frobenius norm, this already solves the initial matrix problem. Its explicit slice is not a discovery.
Máñez-Espina et al. (2026) [5, 6]	Two-mode temporal coupled-mode theory for reciprocal bianisotropic metasurfaces. In the balanced, equal-frequency specialization, the precursor's Eq. (18) gives $\Delta A(\omega_\pm) = \mp r_b g^2/(\gamma^2+g^2)$, for positive coupling convention.	r_b is a background reflection amplitude, not the total response's reflection or transmitted power. This model-specific bound is not an all-profile finite-dielectric optimum.

1.2 Additional results that restrict any novelty claim

Rollett's two-port invariant [7], in reciprocal scattering notation, is

$$K_{\text{R}} = \frac{1-|r_L|^2-|r_R|^2+|\det S|^2}{2|t|^2} = 1+\frac{\det(I-S^\dagger S)}{2|t|^2}. \tag{1}$$

Consequently, the quantity $c = \det(I-S^\dagger S)/|t|^2$, including its invariance under independent lossless coatings, is established. We do not extend this invariance to arbitrary feedback embeddings.

Ma, Pestourie and Johnson [8], Eq. (35), already derive the weak-loss relation

$$\frac{|A_L-A_R|}{A_L+A_R} \leq |r_{\text{lossless}}| + O(\text{loss}).$$

Their perturbative hypothesis differs from the exact finite-loss constraints used below. Guo and Fan [9] provide complete abstract eigenvalue feasibility, and Li et al. [10] characterize coherent control for a given system. Kuang, Zhang and Miller [11] already establish material-dependent single-frequency absorption and thickness limits. Neither positivity of loss nor the general idea of a material-dependent thickness limit can be claimed as new.

Weighted observability has further close precedents. Chitour and Sigalotti [12], Lemma 2.5, prove positive Gramian floors for measurable observation of fixed oscillators. Hante, Sigalotti and Tucsnak [13], Proposition 5.4, establish quantitative small-window observability for a fixed

skew-symmetric generator and controllable observation. More explicitly, Privat, Trélat and Zuazua [14], Remark 2, give, for every measurable $\omega \subset [0,\pi]$ with $|\omega| = \rho\pi$,

$$\frac{\rho\pi - \sin(\rho\pi)}{2} \le \inf_{j\ge 1} \int_\omega \sin^2(jx)\,\mathrm{d}x \le \frac{\rho\pi + \sin(\rho\pi)}{2}.$$

This concerns worst-mode observation for a fixed homogeneous real wave operator. These results preempt a claim that the Gramian method or the local sinusoidal optimization is itself new. The spatial complex Helmholtz generator below is variable and has an indefinite flux form; identifying it with a fixed dissipative time generator does not directly transfer those theorems.

Likewise, the optical Borrmann effect already places thin absorbing layers near standing-wave minima [15, 16]. The construction below uses that established physical mechanism. Its claimed research content is restricted to a uniform all-profile bound and a matching *exact-target* leading coefficient, not a new multilayer architecture.

Priority verdict. No equivalent statement of Theorem 5.1 was found in the retrieved optics, two-port, transmission-bound, or observability sources. This is a precise candidate contribution, but the search does not certify priority. Its classification remains **CONDITIONAL**. Search coverage extends through the actual date, 9 October 2026.

The 2014 main theorem and proof were inspected in the publisher-provided first-page preview; the remaining pages were not retrieved. For the 2026 metasurface paper, published prose and the complete arXiv precursor were read; the publisher PDF and supplement were blocked. Older complex-potential bounds of Martin [17] were identified, but complete primary-text retrieval remained incomplete. These are explicit limitations of the novelty audit, rather than grounds to infer absence of prior work.

2 The established two-port optimization

The optimization in this section is an established consequence of passive scattering-matrix feasibility. In particular, Guo and Fan's necessary and sufficient reflection conditions at prescribed singular values [4] imply it by elimination. We supply an elementary proof, including phases and degeneracies, to specify the benchmark precisely; none of the results in this section is claimed as new.

Let the two ports be power normalized and write

$$S = \begin{pmatrix} r_L & t \\ t & r_R \end{pmatrix}, \qquad \mathcal{A} = I - S^\dagger S \succeq 0, \qquad T = |t|^2. \tag{2}$$

Define

$$\begin{gathered} A_L = 1 - T - |r_L|^2, \quad A_R = 1 - T - |r_R|^2, \\ a = \frac{A_L + A_R}{2}, \quad \delta = \frac{A_L - A_R}{2}, \quad x = |\delta|, \quad b = 1 - T - a. \end{gathered} \tag{3}$$

Thus $A_{L,R} = a \pm \delta$ and $|r_{L,R}|^2 = b \mp \delta$.

Proposition 2.1 (Established feasibility conditions)**.** *Real numbers (T, a, δ) are realized by a symmetric contraction if and only if*

$$0 \le T \le 1, \qquad 0 \le a \le 1 - T, \qquad 0 \le x \le \min(a, b), \qquad F(x) \ge 0, \tag{4}$$

where

$$F(x) = a^2 - x^2 - 2T\big(b - \sqrt{b^2 - x^2}\big). \tag{5}$$

Every pair $0 \le T \le 1$, $0 \le a \le 1-T$ is feasible. For prescribed nonzero t, r_L, r_R, set $\theta = \arg r_L + \arg r_R - 2\arg t$. In addition to the first three conditions in (4), the exact phase condition is

$$\cos\theta \le \frac{a^2 - x^2 - 2Tb}{2T\sqrt{b^2 - x^2}}. \tag{6}$$

If $T|r_L r_R| = 0$, no phase condition is imposed; the remaining determinant condition is $a^2 - x^2 - 2Tb \ge 0$.

Proof. Put $z = \overline{r_L}t + \overline{t}r_R$. Direct multiplication gives

$$\mathcal{A} = \begin{pmatrix} a+\delta & -z \\ -\overline{z} & a-\delta \end{pmatrix}, \qquad \det\mathcal{A} = a^2 - x^2 - |z|^2. \tag{7}$$

A Hermitian two-by-two matrix is positive semidefinite exactly when its diagonal entries and determinant are nonnegative. The reflection magnitudes and diagonal entries give the first three conditions in (4). When all entries are nonzero,

$$|z|^2 = 2T(b + \sqrt{b^2 - x^2}\cos\theta),$$

whose minimum is $2T(b - \sqrt{b^2 - x^2})$. This proves necessity, the phase criterion, and its stated degenerate version. Conversely, the real symmetric matrix

$$S_0 = \begin{pmatrix} \sqrt{b-\delta} & \sqrt{T} \\ \sqrt{T} & -\sqrt{b+\delta} \end{pmatrix} \tag{8}$$

has nonnegative absorption diagonal and determinant $F(x)$, proving sufficiency. Taking $\delta = 0$ gives $S_0^\dagger S_0 = (1-a)I$, proving the assertion about feasible pairs. □

Proposition 2.2 (Established sharp envelope)**.** *For a feasible pair (T, a), put $a_c(T) = (1-T^2)/2$. Then*

$$\max|A_L - A_R| = \begin{cases} 2\sqrt{b^2 - \left(\sqrt{1-2a} - T\right)^2}, & 0 \le a < a_c(T), \\ 2b, & a_c(T) \le a \le 1-T. \end{cases} \tag{9}$$

The formulas agree at the interface and include $T = 0$ and $T = 1$.

Proof. Writing $y = \sqrt{b^2 - x^2}$ yields the identity

$$F(x) = (y+T)^2 - (1-2a). \tag{10}$$

If $a \ge a_c(T)$, then $1 - 2a \le T^2$ and $b \le a$. Consequently $x = b$ is feasible and is maximal. If $a < a_c(T)$, put $y_0 = \sqrt{1-2a} - T > 0$. Equation (10) is nonnegative exactly when $y \ge y_0$. Since $F(0) = a^2 \ge 0$, we have $b \ge y_0$ and may set $x_* = \sqrt{b^2 - y_0^2}$. The equality $F(x_*) = 0$ implies $a^2 - x_*^2 = 2T(b - y_0) \ge 0$; hence $x_* \le a$. The matrix (8) attains this largest feasible x. □

Complete equality characterization. Port interchange is included by allowing either sign of δ below; phases of zero entries are never assigned.

1. If $T = 0$, every optimizer is diagonal and satisfies

$$|\delta| = \min(a, 1-a), \qquad |r_L| = \sqrt{1-a-\delta}, \qquad |r_R| = \sqrt{1-a+\delta}.$$

All phases of nonzero reflections are arbitrary. For $a \le 1/2$ one absorptance vanishes; for $a \ge 1/2$ one reflection vanishes. In particular, $a = 0$ permits arbitrary unit-modulus diagonal entries, $a = 1/2$ gives one zero and one unit-modulus reflection, and $a = 1$ gives $S = 0$.

2. If $T > 0$ and $0 \le a < a_c(T)$, every optimizer has $\delta = \pm x_*$, the magnitudes in (8), and

$$\arg r_L + \arg r_R - 2\arg t = \pi \pmod{2\pi}. \tag{11}$$

All three entries are nonzero. Necessity follows because the largest possible absorption determinant at x_* is already $F(x_*) = 0$; therefore the minimizing phase is mandatory. These conditions are also sufficient. Equivalently, all optimizers are DS_0D, with D diagonal unitary. Their spectra satisfy

$$\operatorname{spec}\mathcal{A} = \{0, 2a\}, \qquad \sigma(S) = \{1, \sqrt{1-2a}\}. \tag{12}$$

For $a > 0$ there is exactly one lossless coherent input direction. For $a = 0$, the maximum is zero, $\mathcal{A} = 0$, and S is unitary; the same phase condition still applies.

3. If $T > 0$ and $a_c(T) \le a \le 1 - T$, every optimizer has $|\delta| = b$, so one reflection is zero and the other has magnitude $\sqrt{2b}$. All phases of nonzero entries are arbitrary, since

$$\det\mathcal{A} = 2a - 1 + T^2 \ge 0. \tag{13}$$

At $a = a_c(T)$ and $T < 1$, the singular values are $1, T$; above that interface $\mathcal{A}$ is positive definite. When $b = 0$, both reflections vanish and only the phase of t is free. The endpoint $T = 1, a = 0$ is therefore a perfect lossless transmitter with arbitrary transmission phase.

Explicit elimination from Guo–Fan. For ordered reflection magnitudes $p \ge q \ge 0$ and singular values $\sigma_1 \ge \sigma_2 \ge 0$, Eqs. (24)–(25) of [4] specialize to

$$p - q \le \sigma_1 - \sigma_2 \le p + q \le \sigma_1 + \sigma_2. \tag{14}$$

The Frobenius identity fixes $\sigma_1^2 + \sigma_2^2 = p^2 + q^2 + 2T$. Put $d = \sigma_1 - \sigma_2$, $c = \sigma_1 + \sigma_2$. At this fixed sum of squares, $(c + d)/2 = \sigma_1$ is nondecreasing in d on $0 \le d \le c$. Its smallest allowed value occurs at $d = p - q$, with $c = \sqrt{(p+q)^2 + 4T}$, which satisfies every inequality in (14). Thus

$$\min\sigma_1 = \frac{p - q + \sqrt{(p+q)^2 + 4T}}{2}. \tag{15}$$

Requiring this norm to be at most one is equivalent to

$$0 \le q \le p \le 1, \qquad 0 \le T \le (1-p)(1+q). \tag{16}$$

Substituting $p = \sqrt{b + x}$ and $q = \sqrt{b - x}$ recovers (4) and its sharp fixed-mean slice (9). Thus the matrix optimization is directly implied by previously complete reflection feasibility, independently of whether its particular piecewise parametrization was written explicitly before.

3 The finite-dielectric problem

Fix a real vacuum wavenumber $k > 0$. Use time dependence $e^{-i\omega t}$, scalar relative permittivity $\varepsilon(z) = p(z) + iq(z)$, relative permeability one, normal incidence, and identical vacuum exterior media. A finite multilayer consists of finitely many homogeneous layers with finite coefficients and nonnegative thicknesses; zero-thickness layers may be discarded. The electric field satisfies the exact equation

$$E'' + k^2\varepsilon(z)E = 0. \tag{17}$$

Electric field and its derivative are continuous at interfaces. No effective-medium, weak-loss, resonant-mode, or thin-film approximation is used in the lower bounds.

For the sharp theorem impose

$$1 \le p(z) \le P, \qquad 0 \le q(z) \le Q, \qquad \int q(z)\,\mathrm{d}z = M, \quad P > 1, \quad Q, M > 0. \tag{18}$$

The total geometric thickness includes lossless spacers and matching layers. There is no prescribed layer count or positive minimum feature size. Here M has dimensions of length and is an integrated imaginary permittivity at the specified frequency; it is not absorbed power or a frequency-integrated material sum rule. The class allows continuously selectable permittivities within the stated caps, rather than a fixed catalogue of measured materials.

3.1 Well-defined scattering and the Gram identity

The outgoing homogeneous problem has no nonzero solution at real k: outward radiated flux plus nonnegative volume dissipation would otherwise sum to zero with a positive term. Finite transfer matrices therefore give a well-defined reciprocal passive scattering matrix. Its transmission amplitude is nonzero. Indeed, zero transmission for a left incident wave would make both E and E' zero at the right boundary, so ODE uniqueness would make the entire field zero, contradicting the unit incident wave.

Let E_L, E_R denote the fields for unit incidence from the left and right. With $v(z) = (E_L(z), E_R(z))$, flux balance gives

$$\mathcal{A} = I - S^\dagger S = k \int q(z) v(z)^\dagger v(z)\,\mathrm{d}z. \tag{19}$$

For any subinterval J, denote the same integral restricted to J by $\mathcal{A}_J$. It is positive semidefinite and $\mathcal{A} - \mathcal{A}_J \succeq 0$.

Let $s(y, x)$ be the fundamental solution of (17) with $s(x, x) = 0$ and $\partial_y s(x, x) = 1$. Constancy of the Wronskian gives

$$E_L(x)E_R'(x) - E_L'(x)E_R(x) = -2ikt,$$

and hence

$$E_L(x)E_R(y) - E_R(x)E_L(y) = -2ikt\, s(y, x).$$

Expanding the determinant of the two-dimensional Gram integral now proves the exact identity

$$\boxed{\frac{\det \mathcal{A}_J}{T} = 4k^4 \int_{\substack{x<y \\ x,y\in J}} q(x)q(y)|s(y, x)|^2\,\mathrm{d}x\,\mathrm{d}y.} \tag{20}$$

The denominator is the transmission of the complete stack, including surroundings. In particular, positive loss on any interval makes $\mathcal{A}$ positive definite: a zero quadratic form in (19) would force a nonzero incident combination to vanish on that interval, contradicting uniqueness. Thus no finite stack with positive distributed loss attains the rank-one active-branch matrix optimum. This qualitative observation alone gives no uniform finite-thickness gap.

3.2 Transfer form and the established coating invariant

For the field state $u = (E, E'/(ik))^{\mathsf{T}}$, let F transfer the left boundary to the right boundary. It has determinant one. Put $\Sigma = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. The normalized directed flux is $u^\dagger \Sigma u/2$. The matrix mapping incident amplitudes to the left field state is

$$B_{\mathrm{in}} = \begin{pmatrix} 1 + r_L & t \\ 1 - r_L & -t \end{pmatrix}, \qquad \det B_{\mathrm{in}} = -2t.$$

Consequently

$$\mathcal{A} = \tfrac{1}{2} B_{\mathrm{in}}^{\dagger}(\Sigma - F^{\dagger}\Sigma F)B_{\mathrm{in}},$$

and

$$c := \frac{\det \mathcal{A}}{T} = \det(\Sigma - F^{\dagger}\Sigma F). \tag{21}$$

If F is replaced by $F_R F F_L$, with $F_j^{\dagger}\Sigma F_j = \Sigma$ and $\det F_j = 1$, the determinant is unchanged. This directly verifies the coating invariance used below, while (1) identifies its classical provenance.

4 A uniform material-resource lower bound

For this section it suffices to have $p \le P$, $0 \le q \le Q$ on a selected interval J of length $L > 0$, with $P, Q > 0$ and $\int_J q = M > 0$. Additional finite passive layers outside J are allowed. Define

$$K = k\sqrt{P}, \qquad f(u) = u^2 - \sin^2 u, \qquad g(u) = \sqrt{f(u)}. \tag{22}$$

Lemma 4.1 (Complex modulus comparison)**.** *For $0 \le h \le \pi/K$ with $[x, x+h] \subseteq J$,*

$$|s(x+h, x)| \ge \frac{\sin(Kh)}{K}. \tag{23}$$

Proof. A zero at distance $h < \pi/K$, together with integration by parts and real parts of (17), would imply

$$\int_x^{x+h} |s'|^2 = k^2 \int_x^{x+h} p|s|^2 \le K^2 \int_x^{x+h} |s|^2.$$

Dirichlet Poincaré gives the incompatible lower bound $(\pi/h)^2 \int |s|^2$. Thus $\rho = |s|$ is positive before π/K. Writing $j = \mathrm{Im}(\bar{s}s')$, direct differentiation away from the initial zero gives

$$\rho'' + K^2\rho = (K^2 - k^2 p)\rho + \frac{j^2}{\rho^3} \ge 0, \qquad \rho(0) = 0, \quad \rho'(0) = 1.$$

Near the origin $\rho(h) = h + O(h^3)$, so the initial conditions and the integrability needed for Duhamel's formula are valid. The kernel $\sin(K(h-v))/K$ is nonnegative for $0 \le v \le h \le \pi/K$. Duhamel's formula proves the claim on the open interval, and continuity gives the endpoint. Interface continuity justifies the same argument for piecewise constant coefficients. □

Lemma 4.2 (One-period capped Fourier moment)**.** *If $0 \le q \le Q$ on an interval of length at most π/K, and $m = \int q$, then*

$$\left| \int q(z) e^{2iKz} \, \mathrm{d}z \right| \le \frac{Q}{K} \sin\!\left(\frac{Km}{Q}\right). \tag{24}$$

Proof. Extend q by zero to one period and set $\theta = 2Kz$. For a chosen phase ϕ, maximizing $\int q \cos(\theta - \phi)\, \mathrm{d}\theta$ at fixed mass fills the arc where the cosine is largest at density Q. Indeed, if q_* is this saturated arc and c_* is its endpoint cosine, then $(q_* - q)(\cos(\theta - \phi) - c_*) \ge 0$ pointwise, and the masses of q_*, q agree. Integrating proves optimality. The arc has angular half-width Km/Q; its moment is $Q\sin(Km/Q)/K$. Choose ϕ to align with the original complex moment. The arc may wrap across the period boundary; allowing this only enlarges the feasible class. This standard mass-allocation argument is included to fix constants, not as a claim of a new moment inequality. □

Lemma 4.3 (Cell determinant and combining cells)**.** *For a cell J_j of length at most π/K, with loss mass m_j,*

$$\frac{\det \mathcal{A}_{J_j}}{T} \geq \frac{Q^2}{P^2} f(u_j), \qquad u_j = \frac{K m_j}{Q} \in [0, \pi]. \tag{25}$$

For any disjoint such cells in J,

$$\sqrt{c} \geq \frac{Q}{P} \sum_j g(u_j). \tag{26}$$

Proof. The comparison and the identity

$$\int_{x<y} q(x)q(y) \sin^2(K(y-x))\,\mathrm{d}x\,\mathrm{d}y = \frac{1}{4}\left[m_j^2 - \left|\int q(x)e^{2iKx}\,\mathrm{d}x\right|^2\right]$$

inserted in (20) prove (25). For positive semidefinite 2×2 matrices A, B, $\sqrt{\det(A+B)} \geq \sqrt{\det A} + \sqrt{\det B}$. For $A \succ 0$ this follows from $\mathrm{tr}(A^{-1/2}BA^{-1/2}) \geq 2\sqrt{\det(A^{-1/2}BA^{-1/2})}$; continuity covers singular A. Repeated application, including any additional absorption outside the selected cells, proves (26). □

Lemma 4.4 (Convexity and a global supporting tangent)**.** *The function g is convex on $[0, \pi/2]$. A tangent at any $0 < x \leq 1/3$ lies below g on all of $[0, \pi]$.*

Proof. For $u > 0$,

$$g''(u) = \frac{H(u)}{f(u)^{3/2}}, \quad H(u) = 2\sin^2 u\, f(u) - (u - \sin u \cos u)^2, \quad H'(u) = 2f(u)\sin(2u).$$

Since $H(0) = 0$, convexity holds on the stated half interval; also $g(0) = g'(0) = 0$ by $f(u) = u^4/3 + O(u^6)$. The identity

$$f(u) = 4\int_0^u (u - v)\sin^2 v\,\mathrm{d}v$$

and the chord inequality $\sin v \geq 2v/\pi$ imply $f(u) \geq 4u^4/(3\pi^2)$ there. Moreover $f'(u) = 4\int_0^u \sin^2 v\,\mathrm{d}v \leq 4u^3/3$. Thus $g'(x) \leq \pi x/\sqrt{3}$. Convexity gives the nonpositive tangent intercept $g(x) - xg'(x) \leq 0$. For $v \in [\pi/2, \pi]$,

$$\frac{g(v)}{v} \geq \sqrt{1 - \frac{4}{\pi^2}} > \frac{\pi}{3\sqrt{3}} \geq g'(x).$$

The tangent is therefore below g on this second interval as well. In particular, using Jensen indiscriminately on all of $[0, \pi]$ would be incorrect; g is not convex on that entire interval. □

Theorem 4.5 (Finite resource floor)**.** *Let $U = KM/Q$, and for positive integers n set*

$$\beta_n = \frac{n^2 Q^2}{P^2} f(U/n).$$

Then, with $n_h = \max\{1, \lceil 2KL/\pi \rceil\}$,

$$\boxed{c \geq \beta_{n_h} \geq \frac{4k^4 M^4}{3Q^2(\pi + 2k\sqrt{P}\,L)^2} > 0.} \tag{27}$$

With $n_f = \max\{1, \lceil KL/\pi \rceil\}$, the stronger bound

$$\boxed{c \geq \beta_{n_f}} \tag{28}$$

holds if $U \le \pi/2$ or $U/n_f \le 1/3$. In particular, for fixed k, P, Q, M, (28) holds for every sufficiently large L, and uniformly over the admissible stacks

$$\liminf_{L\to\infty} (kL)^2 c \ge \frac{\pi^2 (kM)^4}{3PQ^2}. \tag{29}$$

Precisely, every sequence of admissible stacks with budgets $L_j \to \infty$ and the same k, P, Q, M obeys this lower limit with $L = L_j$.

Proof. Partition J into n_h equal cells. Every $u_j \le \pi/2$, so (26) and ordinary Jensen give $c \ge \beta_{n_h}$. The lower estimate on f in Lemma 4.4 yields

$$\beta_{n_h} \ge \frac{4k^4 M^4}{3\pi^2 Q^2 n_h^2}.$$

Use $n_h \le 1 + 2KL/\pi$ to obtain (27). For n_f, the cells have length at most π/K. If $U \le \pi/2$, every u_j still lies in the convexity interval. Otherwise, when $U/n_f \le 1/3$, sum the global supporting tangent at the common mean U/n_f. In either case $\sum g(u_j) \ge n_f g(U/n_f)$. Finally $n_f = KL/\pi + O(1)$ and $\beta_{n_f} = k^4 M^4/(3Q^2 n_f^2) + O(n_f^{-4})$, which proves (29). A stack shorter than L can be enclosed in an interval of length L by adding zero-loss vacuum, so the same uniform bound applies under a thickness cap. □

Remark 4.6 (Strictness and physical equality). For positive distributed loss these finite resource floors are not attained. Inside any positive-loss layer,

$$\mathrm{Im}(\bar{s}s')(z) = -k^2 \int_x^z q(v)|s(v,x)|^2 \,\mathrm{d}v$$

is nonzero after positive propagation. The forcing in the modulus equation is then positive, making (23) strict on a positive-measure set of loss pairs within a cell. At least one cell has such pairs. This already makes the resulting determinant bound strict. No characterization of a finite-L minimizing profile is asserted.

5 Sharp asymptotic thickness at exact fixed response

Fix a target in the strict active branch:

$$0 < T < 1, \qquad 0 < a < \frac{1 - T^2}{2}. \tag{30}$$

Define

$$b = 1 - T - a, \quad s = \sqrt{1 - 2a}, \quad y_0 = s - T, \quad x_0 = \sqrt{b^2 - y_0^2}, \quad D_0 = 2x_0, \quad \alpha = \frac{T y_0}{s x_0}. \tag{31}$$

Here y_0, x_0, α are strictly positive, and D_0 is the established matrix optimum. Let $\mathcal{C}_L$ be the finite multilayers satisfying (18), total thickness at most L, transmitted power exactly T, and mean absorptance exactly a. Define

$$D_*(L) = \sup_{S \in \mathcal{C}_L} |A_L - A_R|$$

when the class is nonempty.

Theorem 5.1 (Sharp exact-target asymptotic)**.** *For fixed $k > 0$, $P > 1$, $Q, M > 0$, and targets (30), $\mathcal{C}_L$ is nonempty for every sufficiently large L, and*

$$\boxed{\lim_{L\to\infty} (kL)^2[D_0 - D_*(L)] = \frac{\alpha\pi^2(kM)^4}{3PQ^2}.} \tag{32}$$

Every finite admissible stack has $D < D_0$. The limit is sharp within the stated ordinary dielectric class, with exact fixed T, a, M.

This is the candidate contribution whose priority remains conditional. Its proof below is an ordinary mathematical proof, separately checked by two analytical reviewers. Numerical and symbolic checks in Section 6 are additional verification.

5.1 Uniform lower bound on the deficit

Write $x = |A_L - A_R|/2$ and

$$F(x) = a^2 - x^2 - 2T\left(b - \sqrt{b^2 - x^2}\right).$$

For every passive matrix, the phase optimization from Section 2 gives $Tc = \det \mathcal{A} \le F(x)$, and $x \le x_0 < b$. Moreover,

$$F(x_0) = 0, \quad F'(x) = -2x\left(1 + \frac{T}{\sqrt{b^2 - x^2}}\right), \quad F''(x) = -2 - \frac{2Tb^2}{(b^2 - x^2)^{3/2}} < 0.$$

The tangent inequality for the concave function F gives

$$F(x) \le [-F'(x_0)](x_0 - x) = \frac{sx_0}{y_0}(D_0 - 2x).$$

Thus the global, not merely local, inequality is

$$\boxed{D_0 - D \ge \alpha c.} \tag{33}$$

Combine this with the full-cell floor (29) to prove the lower limiting constant in (32), uniformly over $\mathcal{C}_L$.

There is also an explicit finite-thickness consequence. If β is any valid resource floor from Theorem 4.5, feasibility requires $a^2 \ge T\beta$, and

$$D \le 2\sqrt{b^2 - \left(\sqrt{1 - 2a + T\beta} - T\right)^2}. \tag{34}$$

This follows by solving $F(x) \ge T\beta$. It is an elementary determinant-constrained matrix envelope, not a separate new matrix theorem and not a claimed sharp finite-thickness Maxwell optimum.

5.2 A realizable sequence with the matching coefficient

Put $g_0 = kM$. For a positive integer m, use m cells of common physical length

$$d = \frac{\pi}{k\sqrt{P}}.$$

Each cell contains a centered lossy slice with $\varepsilon = P + iQ$, width $w = M/(Qm)$, and lossless real-P material on both sides. For sufficiently large m, $0 < w < d$. The integral of q is exactly

M, and all material caps hold. This arrangement is a nodal-loss/Borrmann construction, an established physical principle [15, 16].

To compute it exactly, use dimensionless distance kz and the matrices

$$A = i\begin{pmatrix} 0 & 1 \\ P & 0 \end{pmatrix}, \quad B = -QE_{21}, \quad R = \exp\!\left(\frac{\pi A}{2\sqrt{P}}\right), \quad R^2 = -I.$$

Here E_{21} has its sole nonzero entry in row 2, column 1. With $\xi = kw = g_0/(Qm)$, define the algebraic factor

$$D(\xi) = e^{-A\xi/2}e^{(A+B)\xi}e^{-A\xi/2}. \tag{35}$$

The *physical* cell transfer is $C = RD(\xi)R$, because its spacers have positive dimensionless length $(\pi/\sqrt{P} - \xi)/2$. Append one ordinary real-P quarter-wave layer at each end. Then

$$RC^mR = (-1)^{m-1}D(\xi)^m.$$

The sign only changes transmission phase, so the stripped core may be taken as

$$F_m = D(g_0/(Qm))^m.$$

The negative exponents in (35) are an algebraic factorization; no negative-thickness layer is used. The logarithm is analytic near $\xi = 0$. Since $D(-\xi) = D(\xi)^{-1}$, it is odd, and direct multiplication of the exponential series gives

$$\log D(\xi) = \xi B + \xi^3 C_3 + O(\xi^5), \qquad C_3 = \frac{1}{12}\begin{pmatrix} 0 & -Q \\ Q(P+2iQ) & 0 \end{pmatrix}. \tag{36}$$

For completeness the cubic term is also $[A,[A,B]]/24 + [B,[A,B]]/12$, which evaluates to the displayed matrix. At fixed parameters,

$$m\log D(g_0/(Qm)) = G + m^{-2}H + O(m^{-4}), \quad G = -g_0E_{21}, \quad H = \frac{g_0^3}{Q^3}C_3.$$

Since $G^2 = 0$, the Fréchet derivative of the exponential yields

$$F_m = F_0 + m^{-2}J + O(m^{-4}), \quad F_0 = I + G, \quad J = H + \tfrac{1}{2}(GH + HG) + \tfrac{1}{6}GHG. \tag{37}$$

In particular, $J_{12} = -g_0^3/(12Q^2)$. Now

$$\Sigma - F_0^\dagger \Sigma F_0 = \operatorname{diag}(2g_0, 0).$$

The order-m^{-2} term of the lower-right defect entry is $-2\,\mathrm{Re}\,J_{12} = g_0^3/(6Q^2)$. Therefore

$$\boxed{c_m = \det(\Sigma - F_m^\dagger \Sigma F_m) = \frac{g_0^4}{3Q^2m^2} + O(m^{-4}).} \tag{38}$$

This determines the invariant before matching the target powers.

5.3 Exact target matching with bounded-index dielectric layers

Let $p_0 = \sqrt{b - x_0}$, $r_0 = \sqrt{b + x_0}$, and $t_0 = \sqrt{T}$. The chosen real matrix optimizer is

$$S_0 = \begin{pmatrix} p_0 & t_0 \\ t_0 & -r_0 \end{pmatrix}.$$

The active-branch identities imply

$$r_0 - p_0 = 1 - s, \qquad T = (1 - r_0)(1 + p_0).$$

Its field transfer is triangular:

$$F_{\text{target}} = \begin{pmatrix} A_0 & 0 \\ -h_0 & A_0^{-1} \end{pmatrix}, \qquad A_0 = \frac{1 - r_0}{t_0} \in (0, 1), \quad h_0 = \frac{r_0 - p_0}{t_0} > 0.$$

With $\Lambda(s_j) = \operatorname{diag}(s_j, s_j^{-1})$, choose

$$s_{L,0}^2 = \frac{A_0 h_0}{g_0}, \qquad s_{R,0}^2 = \frac{A_0 g_0}{h_0}. \tag{39}$$

Then $\Lambda(s_{R,0}) F_0 \Lambda(s_{L,0}) = F_{\text{target}}$.

For a transformed limiting sheet, put

$$A = s_L s_R, \quad h = g_0 s_L / s_R, \quad v = A + A^{-1}.$$

Its response is exactly

$$T = \frac{4}{(v + h)^2}, \qquad a = \frac{2hv}{(v + h)^2}. \tag{40}$$

The target has $v - h = 2s/t_0 > 0$. The Jacobian determinant of the response with respect to v, h is

$$\det \frac{\partial(T, a)}{\partial(v, h)} = -\frac{16(v - h)}{(v + h)^5} \neq 0.$$

Also $dv/dA = 1 - A^{-2} \neq 0$ at $A_0 < 1$, while

$$\det \frac{\partial(A, h)}{\partial(\log s_L, \log s_R)} = -2Ah \neq 0.$$

The response map is therefore locally nonsingular.

To apply the implicit function theorem without treating the integer m as continuous, set $\tau = 1/m$ and analytically extend

$$\mathcal{F}(\tau) = \exp\left[\tau^{-1} \log D((g_0/Q)\tau)\right]$$

through $\tau = 0$. The expansion (36) makes the extension analytic, with $\mathcal{F}(0) = F_0$ and leading correction $O(\tau^2)$. The real implicit function theorem supplies positive real ratios

$$s_L(m) = s_{L,0} + O(m^{-2}), \qquad s_R(m) = s_{R,0} + O(m^{-2})$$

for which both actual powers T, a are *exactly* the targets for every sufficiently large integer m.

These diagonal transformations are physically implementable within (18). A quarter-wave layer of real refractive index n has transfer $\left(\begin{smallmatrix} 0 & i/n \\ in & 0 \end{smallmatrix}\right)$. Two consecutive such layers with indices n_1, n_2 give

$$-\operatorname{diag}(n_1/n_2, n_2/n_1).$$

For any fixed ratio $s > 0$, choose an integer j with $|\log s|/j < \frac{1}{2} \log P$, and set

$$n_1 = P^{1/4} e^{(\log s)/(2j)}, \qquad n_2 = P^{1/4} e^{-(\log s)/(2j)}.$$

Both indices lie strictly in $(1, \sqrt{P})$. Repeating this pair j times implements $\Lambda(s)$, up to a harmless sign. The strict inequalities persist under the small implicit-function adjustments. Thus a fixed finite number of passive, lossless dielectric layers realizes both matching networks smoothly, with total thickness $O(1/k)$ uniformly in m. No ideal transformer or singular surface is left in the physical stack. Coating invariance preserves (38) exactly.

5.4 Converting the invariant coefficient to the optimal deficit

After removing the harmless transmission sign, the matched scattering matrix converges to S_0 at order m^{-2}. All three limiting entries are nonzero, so its relative phase satisfies $\theta = \pi + O(m^{-2})$. The exact determinant identity is

$$\det \mathcal{A} = F(x) - 2T\sqrt{b^2 - x^2}(1 + \cos\theta).$$

The phase penalty is $O(m^{-4})$. Taylor expansion about x_0, using $F'(x_0) = -2x_0 s/y_0 \neq 0$, and (38) therefore gives

$$D_0 - D_m = \alpha c_m + O(m^{-4}) = \frac{\alpha g_0^4}{3Q^2 m^2} + O(m^{-4}). \tag{41}$$

Including all spacers and matching layers, the physical thickness is

$$kL_m = \frac{\pi m}{\sqrt{P}} + O(1). \tag{42}$$

Choose a dimensionless constant C such that $kL_m - \pi m/\sqrt{P} \leq C$ for all sufficiently large m. For each large budget L, choose

$$m(L) = \left\lfloor \frac{\sqrt{P}(kL - C)}{\pi} \right\rfloor.$$

The corresponding stack fits within L, $m(L)/(kL) \to \sqrt{P}/\pi$, and it realizes the exact targets. Equation (41) proves the upper limiting constant in (32). Together with the uniform lower bound, this proves Theorem 5.1 for all large thickness budgets, not only a subsequence.

Corollary 5.2 (Asymptotic minimum thickness)**.** *Let $L_{\min}(\eta)$ be the infimum of thicknesses of admissible stacks with $D \geq D_0 - \eta$, keeping all material resources and target powers fixed. Then, as $\eta \downarrow 0$,*

$$kL_{\min}(\eta) \sim \frac{\pi(kM)^2}{Q}\sqrt{\frac{\alpha}{3P\eta}}. \tag{43}$$

Proof. The positive finite-thickness floor forces $L_{\min}(\eta) \to \infty$. The two matching asymptotic estimates for $D_0 - D_*(L)$, with arbitrarily small relative slack, can then be inverted. The constructed family supplies the upper estimate even if the supremum at a particular L is not attained. □

There is no finite-stack equality in (32)'s underlying active matrix ceiling; equality here means the optimal asymptotic coefficient of a supremum. The construction uses m lossy slices and $O(m)$ layers after merging adjacent identical spacers. It does not prove a minimum layer count, nor classify every asymptotically extremizing profile.

6 Numerical and independent verification

6.1 A rational benchmark

Choose

$$P = 4, \quad Q = 2, \quad kM = 1, \quad T = \frac{12}{25}, \quad a = \frac{8}{25}.$$

Then

$$b = \frac{1}{5}, \quad s = \frac{3}{5}, \quad y_0 = \frac{3}{25}, \quad x_0 = \frac{4}{25}, \quad D_0 = \frac{8}{25}, \quad \alpha = \frac{3}{5},$$

and the limiting optimizer is

$$S_0 = \begin{pmatrix} 1/5 & 2\sqrt{3}/5 \\ 2\sqrt{3}/5 & -3/5 \end{pmatrix}, \qquad (A_L, A_R) = (12/25, 4/25).$$

Theorem 5.1 becomes

$$\boxed{\lim_{L\to\infty} (kL)^2 \left[\frac{8}{25} - D_*(L)\right] = \frac{\pi^2}{80}.} \tag{44}$$

For any admissible thickness, let $n = \max\{1, \lceil 2kL/\pi \rceil\}$. Here $U = 1 < \pi/2$, so the full-cell floor always applies:

$$c \geq \frac{1}{4}\left[1 - n^2 \sin^2(1/n)\right], \qquad \frac{8}{25} - D \geq \frac{3}{20}\left[1 - n^2 \sin^2(1/n)\right]. \tag{45}$$

The simpler universal consequence is

$$\frac{8}{25} - D \geq \frac{1}{5(\pi + 4kL)^2}.$$

For the construction, the core expansion is exactly

$$F_m = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} + \frac{1}{m^2}\begin{pmatrix} 1/96 & -1/48 \\ (23+24i)/288 & 1/96 \end{pmatrix} + O(m^{-4}).$$

The matching ratios tend to $s_L = 1/\sqrt{3}, s_R = 1$. At these limiting ratios the response Jacobian with respect to $(\log s_L, \log s_R)$ is

$$\begin{pmatrix} 24/125 & 72/125 \\ 36/125 & -12/125 \end{pmatrix}, \qquad \det = -576/3125 \neq 0.$$

Their exact-target corrections satisfy

$$\log(s_L\sqrt{3}) = -\frac{25}{576m^2} + O(m^{-4}), \qquad \log s_R = \frac{13}{576m^2} + O(m^{-4}).$$

Consequently $c_m = 1/(12m^2) + O(m^{-4})$ and $D_0 - D_m = 1/(20m^2) + O(m^{-4})$.

Lossy slices m	$D_0 - D_m$	$m^2(D_0 - D_m)$	$m^2 c_m$	c_m/β_{full}
4	$3.13782636\,10^{-3}$	0.0502052218	0.0826755263	1.000405918
8	$7.82066448\,10^{-4}$	0.0500522527	0.0831687923	1.000106739
16	$1.95363761\,10^{-4}$	0.0500131229	0.0832921928	1.000027016
32	$4.88313325\,10^{-5}$	0.0500032845	0.0833230479	1.000006775
64	$1.22072318\,10^{-5}$	0.0500008214	0.0833307619	1.000001695
Limit		1/20	1/12	1

Table 2: Exact homogeneous-layer computations for the fixed-target family. The final column uses the interval from the first to the last lossy endpoint, with all lossless matching layers outside that interval, as permitted by Theorem 4.5. That interval is not the total stack thickness.

The calculation uses $k = 2\pi$, so lengths are in vacuum wavelengths. For $m = 4, 8, 16, 32, 64$, total thicknesses are approximately 2.643598, 3.644152, 5.644291, 9.644326, and 17.644335 wavelengths. These include all matching layers and an optional half-wave vacuum phase correction. The selected lossy spans are $(m-1)/4 + 1/(4\pi m)$; their $O(1)$ difference from total thickness does not change the limit (44).

Actual dielectric cascade powers differ from the specified T, a by less than 5×10^{-15}, and the physical cascade agrees with the matched field-transfer calculation within 3.5×10^{-15}. These residuals are floating-point checks, not a substitute for the implicit-function proof of exact target satisfaction.

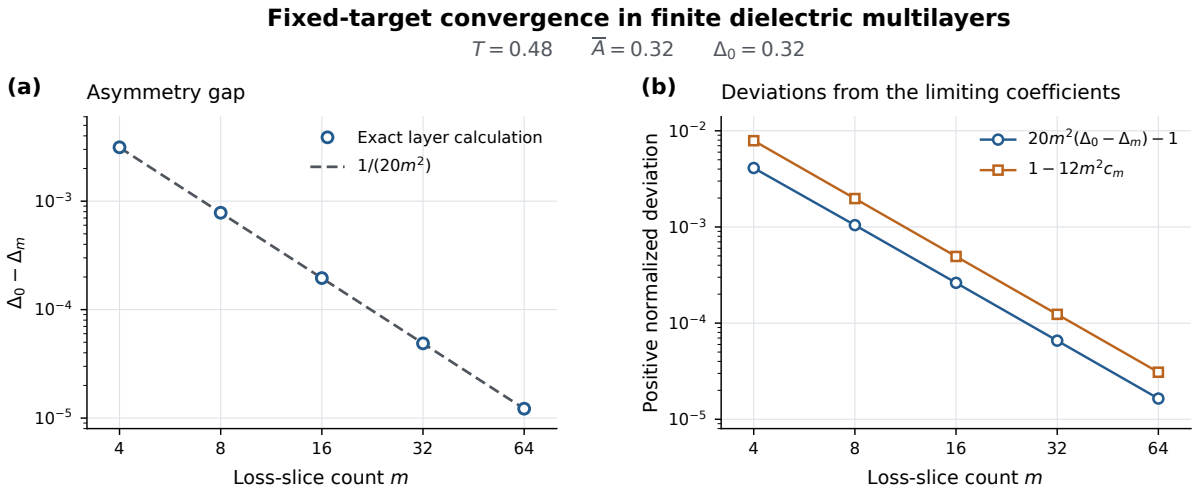


Figure 1: Fixed-target convergence from the existing numerical data. (a) The calculated asymmetry gap follows $1/(20m^2)$. (b) The positive normalized deviations $20m^2(D_0 - D_m) - 1$ and $1 - 12m^2c_m$ approach zero. Here m is the number of lossy slices, not thickness. The homogeneous-layer propagators were checked against an independent scattering cascade.

6.2 Cross-checks, adversarial searches, and limiting cases

The Python/SciPy implementation uses stable scattering cascades, independent interface linear solves, characteristic transfer matrices, and analytic absorption integrals checked by numerical quadrature. In 250 random stacks with one to seven layers, the largest discrepancies were:

Check	Maximum discrepancy
Scattering cascade versus interface solve	8.4×10^{-16}
Characteristic matrix versus interface solve	7.9×10^{-16}
Absorption Gram integral versus $I - S^\dagger S$	1.3×10^{-15}
Analytic absorption integral versus quadrature	1.0×10^{-15}
Lossless unitarity	2.3×10^{-15}
Local determinant identity, relative error	8.8×10^{-14}

Separate SLSQP optimization of the abstract matrix envelope agreed within 8.0×10^{-11}, including both branches and zero-transmission and perfect-transmission limits.

The finite resource floor was challenged with 1200 random profiles, half embedded in additional passive surroundings. None violated the strengthened half-cell floor; the minimum ratio c/β_{n_h} was 1.10173643. A separate set of 2000 complex-ODE comparisons at distances at most $\pi/(2K)$ gave minimum ratio 1.00001279. The extension of the comparison to the full distance π/K is justified by the proof, not inferred from that shorter-distance random sample.

For a uniform layer with $P = 4$, $L = 0.08$ vacuum wavelengths, and $M = QL$, the ratios c/β_1 at $Q = 1, 0.1, 0.01$ were respectively 1.0001588848, 1.0000015888, and 1.0000000159. Thus the short-cell bound has the correct weak-loss limiting coefficient. An eight-cell constrained SLSQP search at $P = 4, Q = 2, L = 0.1, M = 0.1$ found $c = 0.0123243316840$, above the floor 0.0123231683078. This nonconvex optimizer provides an adversarial check; it does not certify the finite-thickness optimum.

6.3 What was independently verified

One analytical agent solved and audited the matrix problem, then checked the full-cell refinement, exact matching, and all-L passage. A second independently reviewed the entire Maxwell argument and the general sharp constant. A separate numerical branch constructed actual dielectric stacks and checked them by independent exact layer formulations. The final LaTeX transcription received a further focused check for changed signs, constants, and assumptions.

The cubic logarithm and the general coefficient $g_0^4/(3Q^2)$ were also verified with exact SymPy polynomial arithmetic. The supplied script checks the general symbolic parameters, not only the rational benchmark. Analytic remainder estimates, uniform quantifiers, and the implicit function theorem remain parts of the written proof.

Lean 4.19.0 was obtained, but its compiler could not start in this environment:

```
error: failed to locate application
```

Its executable self-location required an unavailable `/proc/<pid>/exe` path. The supported `LEAN_SYSROOT` setting did not resolve that startup failure. The accompanying `OQ7Algebra_UNVERIFIED.lean` contains candidate algebraic proofs with no admitted proofs or new axioms, but it has not been compiled. **No Lean theorem is claimed as kernel-verified.** The formalization draft covers envelope identities, phase penalties, determinant-budget algebra, and benchmark inequalities; it does not formalize the Maxwell/ODE theorem.

7 Obstructions, scope, and final verdict

7.1 Failed shortcuts and why they fail

A layer-by-layer bound loses the required information. For a homogeneous slab, write $\sqrt{p+iq} = \nu + i\kappa$, so $q = 2\nu\kappa$. Substituting its exact homogeneous-layer propagator into (21) gives

$$c_{\mathrm{slab}} = \frac{4}{\nu^2+\kappa^2}\left[\nu^2\sinh^2(k\kappa\ell) - \kappa^2\sin^2(k\nu\ell)\right].$$

Expansion at small thickness ℓ gives

$$c_{\mathrm{layer}} = \frac{q^2(k\ell)^4}{3} + O(\ell^6)$$

at fixed material coefficients. Dividing the same material into N artificial sublayers and using only $(\sum_j \sqrt{c_j})^2$ produces a bound proportional to N^{-2}, which collapses under refinement at fixed physical profile. The continuum pair identity retains the cross-location field information that this shortcut discards.

Unrestricted full-period Jensen is false. The function $g(u) = \sqrt{u^2 - \sin^2 u}$ is not convex throughout $[0,\pi]$. The small-mean global tangent in Lemma 4.4 is necessary for the full-period proof at arbitrary fixed total mass.

Loss amount alone gives no positive uniform gap. If the loss-density cap is removed, take a single slab $\varepsilon = P + iq$ of width M/q, and let $q \to \infty$. Its exact transfer approaches $\left(\begin{smallmatrix} 1 & 0 \\ -kM & 1 \end{smallmatrix}\right)$; hence $c \to 0$, while $T \to 4/(2+kM)^2 > 0$. More precisely $c \sim k^4M^4/(3q^2)$. If thickness is unrestricted while P, Q, M remain fixed, the family in Section 5 also has $c \to 0$ and reaches the desired matrix response asymptotically. If the real-permittivity cap is removed, a homogeneous slab with fixed q, ℓ and $p \to \infty$ has $c \sim q^2(k\ell)^2/p \to 0$; this last bare-slab example does not hold arbitrary T, a fixed. These distinguish essential resources from mere passivity.

Positive absorption does not exclude every matrix optimizer. In the high-mean branch, an optimizer with one zero reflection has $\det \mathcal{A} = 2a - 1 + T^2 > 0$ above the interface. The strict gap derived here concerns (30); positive definiteness alone does not exclude all high-mean optimizers.

7.2 What remains unresolved

The exact optimal envelope at every finite L, the complete set of asymptotically optimizing profiles, and a necessary minimum layer count remain open in this investigation. The strict limits $T = 0$, $a = 0$, and $a = (1 - T^2)/2$, as well as the material case $P = 1$, are excluded from Theorem 5.1; its Jacobian or envelope arguments can degenerate there. A bound valid with a fixed positive minimum feature size or a finite catalogue of dispersive materials would be a different realizability problem.

The present result is monochromatic. The hypotheses prescribe material values at one frequency and do not specify dispersion over a frequency interval, so they do not establish a nontrivial finite-bandwidth or causality tradeoff. General vector, oblique-incidence, multipolar, or three-dimensional absorption–scattering asymmetry is likewise not settled.

Verdict	Precise scope
KNOWN	Fixed-T, a passive reciprocal two-port optimum, exact matrix feasibility, and all its equality cases; the Rollett invariant and lossless coating invariance; sinusoidal mass-allocation and nodal-loss architecture.
CONDITIONAL	Theorem 5.1: a proved candidate sharp material-constrained thickness asymptotic with exact fixed response. Two independent reviews accept the mathematics. No equivalent prior theorem was located; publication priority remains unestablished.
UNRESOLVED	General OQ7, finite-bandwidth extensions, exact finite-thickness optimization, and minimum layer count. Lean compilation and kernel verification were not achieved.

The reproducibility archive contains all mathematical review notes, the source of this document and its figure, exact layer calculations, recorded JSON results, the symbolic check, the explicit Lean status, and the literature comparison with retrieval limitations. Downloaded copies of third-party papers are not redistributed in that archive.

References

[1] O. D. Miller and F. Monticone, *Fundamental limits in photonics and electromagnetics: a tutorial* (2026), Section 4.1, OQ7. arXiv:2605.24738.

[2] D. L. Sounas and A. Alù, *Extinction symmetry for reciprocal objects and its implications on cloaking and scattering manipulation*, Optics Letters **39**, 4053–4056 (2014). doi:10.1364/OL.39.004053.

[3] D. L. Sounas and A. Alù, *Time-Reversal Symmetry Bounds on the Electromagnetic Response of Asymmetric Structures*, Physical Review Letters **118**, 154302 (2017). doi:10.1103/PhysRevLett.118.154302.

[4] C. Guo and S. Fan, *Reciprocity Constraints on Reflection*, Physical Review Letters **128**, 256101 (2022), Eqs. (24)–(28). doi:10.1103/PhysRevLett.128.256101. Author-hosted published PDF.

[5] Máñez-Espina, Amrahi, Asadchy, and Díaz-Rubio, *Strong Mode Coupling and Maximum Directional Absorption in Reciprocal Bianisotropic Metasurfaces*, Nanophotonics **15**, e70145 (2026). doi:10.1002/nap2.70145.

[6] Máñez-Espina et al., *Strong Mode Coupling via Quasi-Bound States in the Continuum in Bianisotropic Metasurfaces*, precursor (2025), Eq. (18). arXiv:2512.08927.

[7] J. M. Rollett, *Stability and Power-Gain Invariants of Linear Twoports*, IRE Transactions on Circuit Theory **9**, 29–32 (1962). doi:10.1109/TCT.1962.1086854.

[8] W. Ma, R. Pestourie, and S. G. Johnson, *Time-reversal-symmetry bounds on electromagnetic fields*, Physical Review A **111**, 063518 (2025), Section VII, Eq. (35). doi:10.1103/8bcn-796c. arXiv:2503.03342.

[9] C. Guo and S. Fan, *Passivity constraints on the relations between transmission, reflection, and absorption eigenvalues*, Physical Review B **110**, 205431 (2024). doi:10.1103/PhysRevB.110.205431.

[10] Li, Kim, Fan, and Guo, *Joint control of coherent transmission, reflection, and absorption*, Physical Review B **112**, 245421 (2025). doi:10.1103/nw8t-326d.

[11] Z. Kuang, L. Zhang, and O. D. Miller, *Maximal single-frequency electromagnetic response*, Optica **7**, 1746–1759 (2020). doi:10.1364/OPTICA.398715.

[12] Y. Chitour and M. Sigalotti, *On the Stabilization of Persistently Excited Linear Systems*, SIAM Journal on Control and Optimization **48**, 4032–4055 (2010), Lemma 2.5. doi:10.1137/080737812. arXiv:0810.2122.

[13] F. M. Hante, M. Sigalotti, and M. Tucsnak, *On Conditions for Asymptotic Stability of Dissipative Infinite-Dimensional Systems with Intermittent Damping*, Journal of Differential Equations **252**, 5569–5593 (2012), Proposition 5.4. doi:10.1016/j.jde.2012.01.037. arXiv:1108.4327.

[14] Y. Privat, E. Trélat, and E. Zuazua, *Optimal Observation of the One-dimensional Wave Equation*, Journal of Fourier Analysis and Applications **19**, 514–544 (2013), Remark 2. doi:10.1007/s00041-013-9267-4. Author preprint.

[15] I. V. Kozhevnikov, S. Yulin, T. Feigl, and N. Kaiser, *Effect of anomalous transmittance in EUV multilayer optics*, Optics Communications **281**, 3025–3031 (2008). doi:10.1016/j.optcom.2008.01.057.

[16] J. L. P. Barreaux et al., *Narrowband and tunable anomalous transmission filters for spectral monitoring in the extreme ultraviolet wavelength region*, Optics Express **25**, 1993–2008 (2017), Section 2. doi:10.1364/OE.25.001993.

[17] A. Martin, *Some simple inequalities in scattering by complex potentials*, Il Nuovo Cimento **23**, 641–654 (1962). doi:10.1007/BF02733216. Abstract and indexed primary conclusion inspected; full-text comparison incomplete.